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ADVANCED SYSTEM FOR DRONE DETECTION, TRACKING, AND NEUTRALIZATION

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Abstract

The democratization of aerial maneuverability has exposed a critical vulnerability in modern air defense known as the "Granularity Gap." Conventional Air Surveillance Radars, optimized for high-velocity, large-RCS ballistic targets, are mathematically constrained by the inverse fourth-power attenuation law, rendering them blind to micro-drones ($\sigma < 0.01 \text{ m}^2$) operating within dense urban ground clutter.

This monograph establishes the theoretical architecture and mathematical framework for an autonomous, edge-native Cyber-Physical System designed for drone detection, tracking, and theoretical neutralization. The proposed topology abandons monolithic single-sensor dependency in favor of a multimodal approach. The system incorporates an Omnidirectional Antenna and a 6G-ready Spectrum Analyzer to detect ambient control link anomalies. Upon cueing, the perception layer utilizes an INT8-quantized YOLOv8 convolutional neural network, hosted on a Raspberry Pi 5 computing node, to process visual streams from a high-gain PTZ optical sensor.

To resolve the non-linear kinematics of highly maneuverable targets and compensate for mechanical actuation latency, the architecture mathematically derives and implements an Extended Kalman Filter (EKF). The system's hardware infrastructure encompasses a Display Screen for tactical Human-Machine Interfacing, a high-density LiFePO₄ Battery for autonomous power distribution, and theoretically maps the integration of a Directional Antenna coupled with a Gallium Nitride (GaN) Power Amplifier for active RF jamming. This comprehensive framework offers a rigorous blueprint for neutralizing asymmetric aerial threats using accessible Commercial Off-The-Shelf (COTS) components, establishing the protocols for upcoming empirical field validations.

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List of Abbreviations

Acronym	Description
C-UAS	Counter-Unmanned Aerial Systems
CER	Cost-Exchange Ratio
CIP	Critical Infrastructure Protection
CNN	Convolutional Neural Network
CONOPS	Concept of Operations
COTS	Commercial Off-The-Shelf
EKF	Extended Kalman Filter
EIRP	Effective Isotropic Radiated Power
GaN	Gallium Nitride
HMI	Human-Machine Interface
HUD	Heads-Up Display
KPI	Key Performance Indicator
PA	Power Amplifier
PTZ	Pan-Tilt-Zoom
RCS	Radar Cross Section
YOLO	You Only Look Once
AI	Artificial Intelligence
AJ	Anti-Jamming
ASR	Air Surveillance Radars
BLDC	Brushless DC
C2	Command and Control
CEW	Cognitive Electronic Warfare
CPS	Cyber-Physical System
DANL	Displayed Average Noise Level
DKF	Discrete Kalman Filter
DMA	Direct Memory Access
DSSS	Direct Sequence Spread Spectrum
ECM	Electronic Countermeasures
EO	Electro-Optical
EP	Electronic Protection
FAR	False Alarm Rates
FHSS	Frequency Hopping Spread Spectrum
FPGA	Field Programmable Gate Array
GCS	Ground Control Station
GUI	Graphical User Interface
HIL	Hardware-in-the-Loop

Acronym	Description
HPBW	Half-Power Beamwidth
IF	Intermediate Frequency
IMD	Intermodulation Distortion
IP3	Third-Order Intermodulation Products
ISM	Industrial, Scientific, and Medical
LNA	Low Noise Amplifier
LO	Local Oscillator
LSS	Low-Slow-Small
LWIR	Long-Wave Infrared
MEMS	Micro-Electromechanical Systems
MOT	Multi-Object Tracking
MTI	Moving Target Indicator
MVDR	Minimum Variance Distortionless Response
NPU	Neural Processing Unit
OSD	On-Screen Display
RBW	Resolution Bandwidth
RF	Radio Frequency
SA	Spectrum Analyzer
SADP	Search Active Devices Protocol
SAM	Surface-to-Air Missiles
SCD	Spectral Correlation Density
SITL	Software-in-the-Loop
SNR	Signal-to-Noise Ratio
SoC	System-on-Chip
SOD	Small Object Detection
STT	Single-Target Tracking
UAS	Unmanned Aerial Systems
UKF	Unscented Kalman Filter
UT	Unscented Transform

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Introduction, Problem

1 Formulation, and Theoretical Background

1.1 Introduction: The Proliferation of Asymmetric Aerial Threats

Historically, the doctrine of airspace dominance and Critical Infrastructure Protection (CIP) was predicated on a fundamental assumption: the operational sky was the exclusive domain of state actors operating highly visible, supersonic, and technologically complex platforms. Consequently, global air defense grids were engineered as colossal sentinels designed to detect large Radar Cross Section (RCS) targets, such as intercontinental ballistic missiles, strategic bombers, and conventional tactical aircraft [1, 3].

However, the rapid convergence and commercialization of micro-electromechanical systems (MEMS), high-density Lithium-Polymer (Li-Po) energy storage, and advanced brushless DC (BLDC) motor efficiency have catalyzed a profound paradigm shift. The third decade of the 21st century marks the democratization of aerial maneuverability. The mass proliferation of Class I Unmanned Aerial Systems (UAS)—commonly referred to as micro-drones or loitering munitions—has introduced a devastatingly effective modality of asymmetric security threats [13, 17].

These Commercial Off-The-Shelf (COTS) devices, originally engineered for civilian topography and photography, have been rapidly weaponized. Their defining physical and kinematic characteristics—extremely low procurement cost, hyper-agility exceeding $4g$ instantaneous acceleration, and a minimal electromagnetic footprint ($\sigma < 0.01 m^2$)—render them practically invisible to legacy early-warning radar systems, necessitating a paradigm shift towards multimodal sensing and Edge Artificial Intelligence (AI) [16, 14].

Table 1.1: Comparison between Legacy Aerial Threats and Class I Asymmetric Threats (Micro-Drones).

Characteristic	Legacy Threats (Fighters/Missiles)	Class I UAS (Micro-Drones)
Radar Cross Section (RCS)	Large ($> 1.0 m^2$)	Extremely Small ($< 0.01 m^2$)
Kinematics	Supersonic, High Altitude	Low-Slow-Small (LSS), Hovering
Procurement Cost	\$10M – \$100M+	\$500 – \$2,000 (COTS)
Electromagnetic Signature	High-power military radar/comms	Weak ISM band telemetry (2.4/5.8 GHz)
Detection Vulnerability	Conventional Early-Warning Radar	Multi-modal (RF Sensing + Edge AI)

1.2 Problem Formulation: The Granularity Gap and Economic Asymmetry

The core vulnerabilities exposed by Class I UAS represent a catastrophic intersection of electromagnetic physics constraints and economic asymmetry.

1.2.1 The Granularity Gap and Thermal Submersion in Radar Systems

Legacy surveillance radars rely on the standard Radar Range Equation, where the maximum detection range R_{max} is governed by the fourth root of the target's RCS (σ):

$$R_{max} = \sqrt[4]{\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{min} L_{sys}}} \quad (1.1)$$

For a micro-drone target ($\sigma < 0.01 \text{ m}^2$), the reflected power P_r drops precipitously below the receiver's thermodynamic noise floor (kTB) [2, 5]. The signal is entirely submerged within the ground clutter reflectivity (σ_c^0) [8, 9]. Furthermore, micro-drones executing hovering maneuvers possess a radial velocity $v_r \approx 0$, causing them to fall precisely into the radar's Moving Target Indicator (MTI) Zero-Velocity Notch Filter, essentially blinding the radar network [11].

1.2.2 The Cost-Exchange Ratio (CER) Collapse

The secondary vulnerability is fundamentally economic. Defense economics governing traditional Surface-to-Air Missiles (SAM) operate on a highly unfavorable Cost-Exchange Ratio (CER) when deployed against autonomous micro-drones [19].

$$CER = \frac{C_{intercept}}{C_{target}} \approx \frac{\$3,000,000}{\$1,500} = 2000 : 1 \quad (1.2)$$

Intercepting low-cost swarm assets with multi-million-dollar kinetic interceptors dictates an unsustainable attrition warfare model. Defense mechanisms must therefore leverage directed electromagnetic energy (RF Jamming) and localized Edge AI to bring the cost of interception down to the cost of standard electricity.

1.3 Electromagnetic Foundations: Antenna Theory and Characteristics

To effectively detect RF telemetry and subsequently neutralize the target via jamming, a rigorous understanding of antenna electromagnetics is imperative. An antenna acts as a reciprocal transducer matching the impedance of free space ($Z_0 \approx 377 \Omega$) to the transmission line [4].

1.3.1 Radiation Pattern and Isotropic Radiators

The radiation pattern is a mathematical function or graphical representation of the radiation properties of the antenna as a function of space coordinates (spherical coordinates θ, ϕ). A theoretical *Isotropic*

Radiator radiates power uniformly in all directions (4π steradians). Real antennas, however, focus energy into specific lobes.

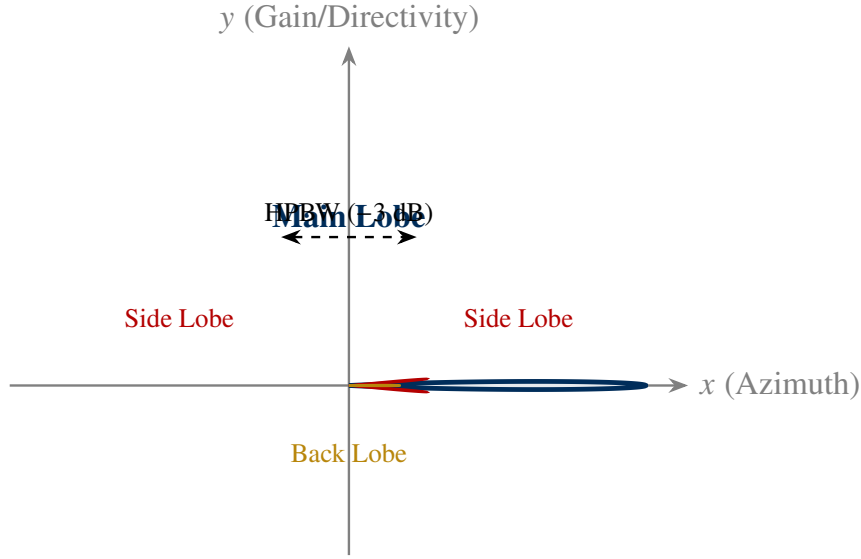


Figure 1.1: 2D Cartesian projection of a directional antenna radiation pattern, illustrating the Main Lobe, Side Lobes, Back Lobe, and the Half-Power Beamwidth (HPBW).

1.3.2 Directivity (D) and Antenna Gain (G)

Directivity (D) is defined as the ratio of the radiation intensity in a given direction to the radiation intensity averaged over all directions [6]. Let $U(\theta, \phi)$ be the radiation intensity (W/sr) and P_{rad} be the total radiated power:

$$D(\theta, \phi) = \frac{U(\theta, \phi)}{U_{avg}} = \frac{4\pi U(\theta, \phi)}{P_{rad}} \quad (1.3)$$

The maximum directivity D_{max} usually occurs at the center of the main lobe.

Antenna Gain (G) is a more comprehensive metric than directivity as it accounts for the ohmic and dielectric losses within the antenna structure. If η_{rad} is the radiation efficiency coefficient ($0 \leq \eta_{rad} \leq 1$), the Gain is mathematically expressed as:

$$G(\theta, \phi) = \eta_{rad} \cdot D(\theta, \phi) \quad (1.4)$$

In the context of the proposed system:

- **Omnidirectional Antennas (Low Gain, 360° coverage):** Utilized in the spectrum sensing phase to detect drone telemetry from any unknown azimuth.
- **Directional Antennas (High Gain, Narrow Beamwidth):** Utilized during the jamming phase to focus maximum Effective Isotropic Radiated Power (EIRP) strictly towards the targeted drone, minimizing collateral interference.

1.4 Spectrum Sensing and the Spectrum Analyzer Architecture

To overcome the radar blind spot, passive RF Spectrum Sensing is employed. Drones continuously emit RF energy for telemetry, video downlinking (e.g., FPV), and C2 (Command and Control) synchronization, primarily within the 2.4 GHz and 5.8 GHz ISM bands [21, 24].

1.4.1 The Superheterodyne Spectrum Analyzer

The core device for capturing these signals is the Spectrum Analyzer (SA). Modern high-frequency SAs (such as the HTOOL-SA-6G) operate on the **Superheterodyne** principle, which translates incoming high-frequency RF signals down to a lower, fixed Intermediate Frequency (IF) for precise filtering and analysis.

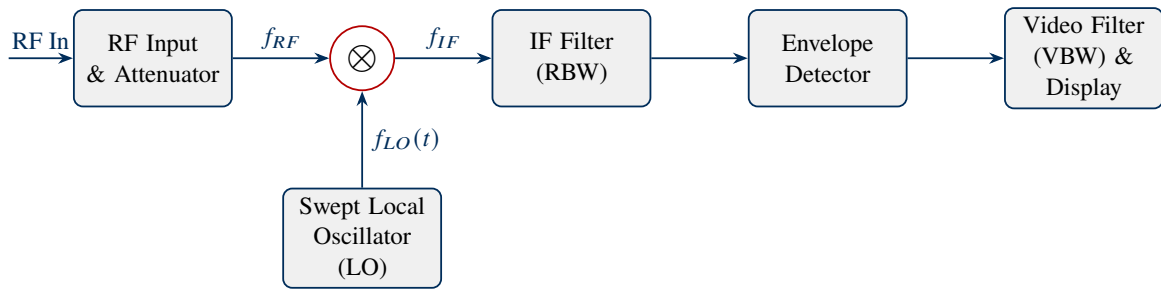


Figure 1.2: Block Diagram of a Superheterodyne Spectrum Analyzer. The incoming f_{RF} is mixed with a swept f_{LO} to produce a constant f_{IF} ($f_{IF} = |f_{RF} - f_{LO}|$), allowing the RBW filter to isolate specific spectral components.

1.4.2 Resolution Bandwidth (RBW) and Displayed Average Noise Level (DANL)

The ability of the system to detect extremely weak drone signals at long ranges is governed by the **Resolution Bandwidth (RBW)** filter. The RBW determines the frequency resolution of the analyzer. A narrower RBW separates closely spaced signals and drastically lowers the noise floor, revealing hidden telemetry spikes.

The absolute sensitivity of the analyzer is mathematically defined by the Displayed Average Noise Level (DANL):

$$DANL \text{ (dBm)} = -174 \text{ dBm/Hz} + 10 \log_{10}(RBW) + NF \quad (1.5)$$

Where -174 dBm/Hz is the thermal noise power at room temperature (kT_0), and NF is the Noise Figure of the internal circuitry [41]. By decreasing the RBW by a factor of 10, the noise floor drops by 10 dB, significantly extending the drone detection horizon.

1.5 Electronic Countermeasures: Jamming and Microwave Integrity

Once a UAS is localized and verified, active neutralization is executed via Electronic Countermeasures (ECM), specifically RF Jamming. The objective is to sever the link between the Ground Control Station (GCS) and the drone's microwave receiver [42].

1.5.1 Mechanics of RF Jamming (The J/S Ratio)

Jamming is achieved by transmitting a high-power noise or structured signal that overwhelms the legitimate C2 signal. The effectiveness is dictated by the **Jamming-to-Signal (J/S) Ratio**. Using the Friis transmission equations, the ratio at the drone's receiver is derived as:

$$\frac{J}{S} = \frac{P_j G_j}{P_s G_s} \left(\frac{R_s}{R_j} \right)^2 \cdot L_p \geq \gamma_{th} \quad (1.6)$$

Where P is power, G is antenna gain, R_s is the distance from the drone to its pilot, R_j is the distance from the jammer to the drone, L_p represents polarization mismatch losses, and γ_{th} is the minimum "Burn-Through" threshold (typically 15 – 20 dB) [43].

1.5.2 Effect of Jamming on Microwave Receivers

When a high-power jamming signal hits the drone's microwave receiver front-end, it directly attacks the **Low Noise Amplifier (LNA)**.

- **Desensitization and Saturation:** Microwave LNAs have a linear operational region bounded by the 1 dB Compression Point (P_{1dB}). When the high-amplitude jamming wave enters the LNA, it drives the transistor into its non-linear saturation region. The gain of the LNA drops precipitously, rendering it deaf to the weak, legitimate control signal.
- **Intermodulation Distortion (IMD):** High-power barrage jamming creates harmonic frequencies. If multiple signals mix within the non-linear LNA, they produce Third-Order Intermodulation Products (IP3) that fall directly into the IF bandpass, completely corrupting the demodulator's bit-error rate (BER) [41].

1.5.3 Anti-Jamming (AJ) Technologies in Modern UAS

To maintain academic rigor, the defense system must account for drones equipped with Electronic Protection (EP) or Anti-Jamming (AJ) capabilities. Modern military and high-end commercial drones employ **Frequency Hopping Spread Spectrum (FHSS)** and **Direct Sequence Spread Spectrum (DSSS)** [42].

In DSSS systems, the narrow-band information is multiplied by a high-speed pseudo-noise (PN) code, spreading the signal over a wide bandwidth. The resilience of the drone to jamming is quantified by the **Processing Gain (G_p)**:

$$G_p (dB) = 10 \log_{10} \left(\frac{B_{spread}}{B_{info}} \right) \quad (1.7)$$

To overcome advanced FHSS/DSSS systems, the proposed architecture advocates for **Follower/Spot Jamming** (dynamically shifting the jamming energy to follow the hopping sequence) or **Adaptive Spatial Beamforming** to physically focus energy rather than spreading it across the spectrum [43].

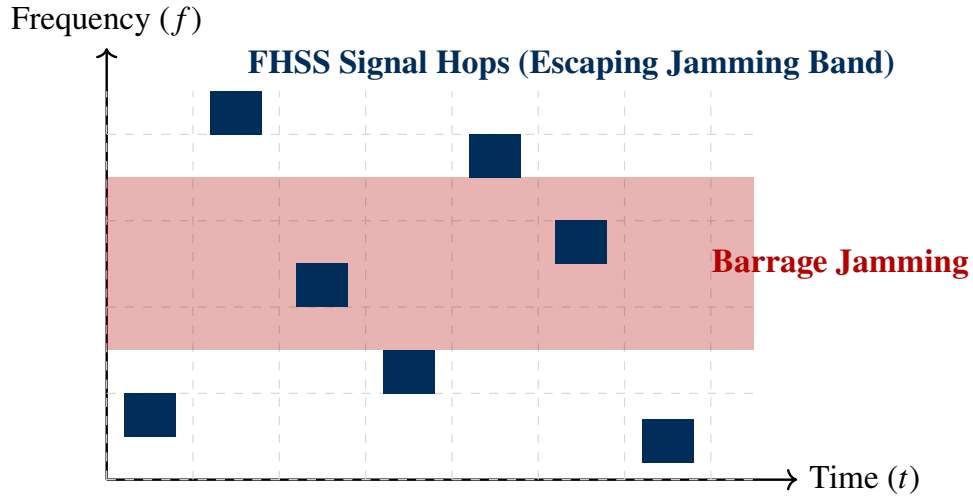


Figure 1.3: Time-Frequency grid illustrating Frequency Hopping Spread Spectrum (FHSS) operating against a Partial-Band Barrage Jammer. The drone rapidly changes its carrier frequency based on a pseudo-random code ($f_k = f_0 + c_k \Delta f$).

1.6 Proposed Cyber-Physical Architecture

To address the aforementioned constraints, this project proposes the design and implementation of an advanced, edge-native Cyber-Physical System (CPS). The core topology abandons monolithic radar dependency in favor of multimodal sensor fusion:

1. **Spectrum Sensing Layer:** Utilizing the HTOOL-SA-6G spectrum analyzer and omnidirectional antennas to monitor the RF noise floor and isolate FHSS/DSSS drone uplinks based on specific frequency signatures [24].
2. **Optical AI Verification:** Upon RF cueing, a high-gain Pan-Tilt-Zoom (PTZ) camera visually acquires the target. An INT8-quantized YOLOv8 architecture [30, 33], running on a Raspberry Pi 5 [45], performs real-time classification to reject biological clutter (birds).
3. **Kinematic Filtering:** An Extended Kalman Filter (EKF) [35] mathematically sustains the target's trajectory vector during occlusions, feeding predictive coordinates to the PTZ motor drivers.
4. **Directional Neutralization:** Theoretical application of High-Gain directional antennas and GaN Power Amplifiers [44] to execute targeted J/S overrides on the drone's LNA.

1.7 Project Objectives

The primary objective is to bridge the "Granularity Gap" in modern air defense. To transition this theoretical framework into a rigorously tested prototype, the system is strictly bound by the following Key Performance Indicators (KPIs):

- **Spectrum Analysis Proficiency:** Accurately configure RBW and VBW filters to lower the DANL, ensuring the detection of Class I drones at ranges $\geq 1.0 \text{ km}$ despite urban RF clutter.

-
- **System Loop Latency Constraint (τ_{sys}):** Ensure the total system latency—from optical acquisition to motor actuation—remains $\leq 200\text{ ms}$ to prevent phase instability in the PID tracking loop [39].
 - **Electromagnetic Optimization:** Mathematically and physically select antennas that maximize Directivity (D) and Gain (G) for the jamming array, validating the capability to induce LNA saturation on commercial microwave receivers.
 - **AI Execution:** Successfully interface high-speed RTSP video streams within a multiprocessing, zero-copy DMA architecture on the Raspberry Pi 5, maintaining $\geq 30\text{ FPS}$ inference without inducing SoC thermal throttling.

2

Problem Statement, Literature Review, and Analysis

2.1 Introduction

The democratization of Unmanned Aerial Systems (UAS) has precipitated a structural crisis in modern air defense doctrines [16]. As the academic literature demonstrates, the transition from monolithic conventional radar systems to decentralized, multimodal Cyber-Physical Systems is an operational necessity [18]. This chapter rigorously defines the physical problem statement underlying the "Granularity Gap," systematically reviews the state-of-the-art across sensing modalities, and critically analyzes the electromagnetic warfare paradigms—specifically RF Spectrum Sensing, Jamming, and Anti-Jamming techniques—that govern the modern Counter-UAS (C-UAS) operational environment.

2.2 Problem Statement: The Physics of the Granularity Gap

The inability of conventional Air Surveillance Radars (ASR) to detect Class I micro-drones is rooted in immutable electromagnetic limitations and statistical detection theories [1, 3].

2.2.1 Thermal Submersion and Swerling Target Models

The fundamental limit of a monostatic radar is dictated by the Radar Range Equation. The maximum theoretical detection range R_{max} is governed by the fourth root of the target's Radar Cross Section (RCS), denoted as σ :

$$R_{max} = \sqrt[4]{\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 (k T_0 B F) L_{sys}}} \quad (2.1)$$

Micro-drones exhibit an RCS of $\sigma < 0.01 \text{ m}^2$. Furthermore, due to the complex geometry of their multi-rotor frames, the RCS is highly aspect-dependent and rapidly fluctuates [5]. This fluctuation is mathematically modeled using the **Swerling Case 1 and 3** models, which describe targets consisting of many independent scatterers. The probability density function (PDF) of the RCS for a Swerling 1 target follows a Rayleigh distribution:

$$p(\sigma) = \frac{1}{\sigma_{avg}} \exp\left(-\frac{\sigma}{\sigma_{avg}}\right), \quad \sigma \geq 0 \quad (2.2)$$

This exponential fluctuation dictates that maintaining a constant Probability of Detection (P_d) requires a disproportionately massive increase in the Signal-to-Noise Ratio (SNR), pushing the required transmit power (P_t) beyond practical embedded limits [2, 7].

2.2.2 Clutter Statistics and the Doppler Blind Zone

In urban topography, the weak target return is actively submerged within thermodynamic ground clutter. The clutter reflectivity (σ_c^0) often obeys a Weibull or K-distribution rather than Gaussian, resulting in "heavy tails" that cause severe False Alarm Rates (FAR) [8, 10].

To mitigate clutter, Pulse-Doppler radars employ Moving Target Indicator (MTI) algorithms, filtering returns based on their Doppler frequency shift ($f_d = \frac{2v_r}{\lambda}$). To eliminate stationary buildings, signal processors apply a "Zero-Velocity Notch Filter". When a micro-drone executes a hover maneuver ($v_r \approx 0$), it falls precisely into this notch. The MTI filter's frequency response $H(f)$ near zero velocity approaches zero, forcing the defense grid to actively discard the drone's signature as static background clutter [11, 12].

2.3 Literature Review: Spectrum Sensing in C-UAS

To circumvent the inherent radar blind spot, modern research has aggressively pivoted towards passive Radio Frequency (RF) Spectrum Sensing [22, 23]. Micro-drones continuously emit RF energy for telemetry and video downlinks within the 2.4 GHz and 5.8 GHz Industrial, Scientific, and Medical (ISM) bands.

2.3.1 Superheterodyne Receiver Architecture

The core hardware enabling high-fidelity RF analysis is the Superheterodyne Spectrum Analyzer. This architecture translates incoming high-frequency RF signals down to a lower, fixed Intermediate Frequency (IF) for precise filtering.

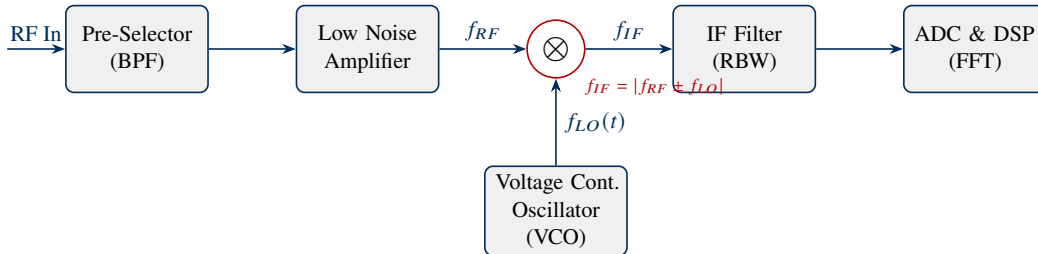


Figure 2.1: Detailed block diagram of a Superheterodyne Spectrum Analyzer front-end. The Resolution Bandwidth (RBW) filter determines the noise floor according to the equation: $DANL = -174 + 10 \log_{10}(RBW) + NF$.

2.3.2 Mathematical Paradigms of Detection Algorithms

Spectrum sensing algorithms implemented within Digital Signal Processors (DSP) generally fall into three distinct mathematical paradigms [24, 21]:

1. **Energy Detection (ED):** The detector measures the energy of the received signal over an observation time T and compares it to a noise threshold (λ_{th}). Let $y(t) = s(t) + n(t)$. The test statistic is:

$$T_{ED} = \frac{1}{N} \sum_{n=1}^N |y(n)|^2 \underset{H_0}{\overset{H_1}{\gtrless}} \lambda_{th} \quad (2.3)$$

While computationally efficient, ED is highly susceptible to Noise Uncertainty and cannot differentiate between a drone signal and generic Wi-Fi interference.

2. **Matched Filter Detection:** If the specific drone protocol (e.g., DJI OcuSync) is known a priori, a matched filter maximizes the SNR. It requires perfect synchronization and demodulation, making it impractical for unknown or proprietary UAS.
3. **Cyclostationary Feature Extraction:** Drone telemetry protocols exhibit inherent periodicities (cyclic frequencies) due to carrier waves and pulse trains. The Spectral Correlation Density (SCD) function is utilized to detect these features even when the signal is deeply buried below the noise floor ($SNR < 0 \text{ dB}$). The SCD is the Fourier transform of the cyclic autocorrelation function $R_y^\alpha(\tau)$:

$$S_y^\alpha(f) = \int_{-\infty}^{\infty} R_y^\alpha(\tau) e^{-j2\pi f\tau} d\tau \quad (2.4)$$

Where α is the cyclic frequency. This method strongly rejects stationary AWGN, isolating the artificial drone signatures.

2.4 Electronic Countermeasures (ECM): Jamming Physics

Upon positive detection, active neutralization is executed via ECM. Jamming is fundamentally an electromagnetic energy supremacy contest at the target's receiver front-end, aiming to sever the drone from its Ground Control Station (GCS) [41].

2.4.1 The Burn-Through Range and J/S Ratio

The effectiveness of jamming is dictated by the **Jamming-to-Signal (J/S) Ratio** [42]. The "Burn-Through Range" is the specific distance at which the legitimate control signal power finally overcomes the jamming noise. By rearranging the Friis transmission equations, the J/S ratio at the drone's receiver is derived as:

$$\frac{J}{S} = \frac{EIRP_{Jammer}}{EIRP_{GCS}} \left(\frac{R_{GCS}}{R_{Jammer}} \right)^2 \cdot L_{pol} \geq \gamma_{th} \quad (2.5)$$

Where γ_{th} is the minimum threshold (typically 15 – 20 dB for standard FSK links).

2.4.2 Impact on Microwave Circuits: IP3 and IMD

A drone's microwave receiver relies on a Low Noise Amplifier (LNA). The jamming array's objective is to physically drive this LNA into non-linear saturation [44].

Microwave amplifiers exhibit linear gain only up to the 1 dB Compression Point (P_{1dB}). When immense RF energy hits the receiver, the LNA saturates (Desensitization), rendering the drone deaf. Furthermore, high-power barrage jamming induces **Intermodulation Distortion (IMD)**. If multiple signals mix within the saturated LNA, they produce Third-Order Intermodulation Products (IP3).

2.5 Electronic Protection (EP): Anti-Jamming Architectures

A robust C-UAS platform must account for advanced UAS equipped with Anti-Jamming (AJ) capabilities. The literature identifies Spread Spectrum and Spatial Filtering as the primary defense mechanisms

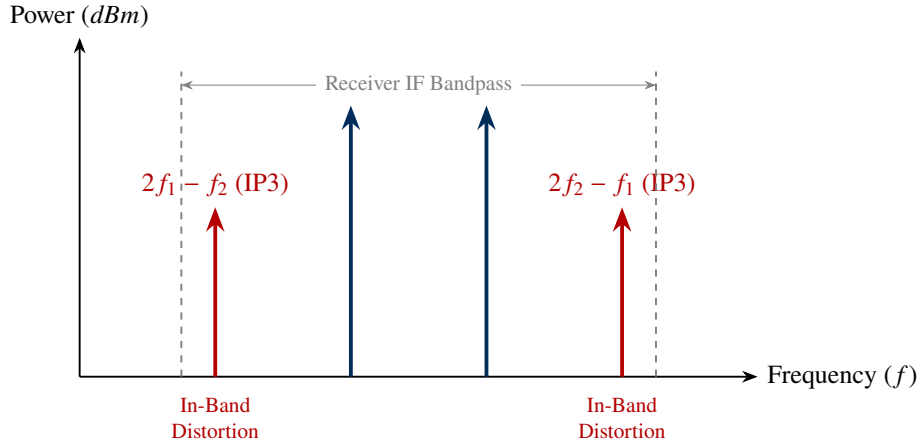


Figure 2.2: Frequency spectrum illustrating Third-Order Intermodulation Distortion (IMD). The high-power jamming signals (f_1, f_2) force the non-linear LNA to generate IP3 products that fall directly into the receiver's IF bandpass, irrecoverably corrupting the data link.

[43, 42].

2.5.1 Frequency Hopping Spread Spectrum (FHSS)

In FHSS, the drone rapidly switches its carrier frequency across the available spectrum based on a pseudo-random code. Against FHSS, a narrow-band "Spot Jammer" fails entirely. The defense system must employ either **Wideband Barrage Jamming** (spreading noise across the entire hopping bandwidth, which drastically dilutes the power spectral density) or a **Follower Jammer** (a Digital RF Memory - DRFM system that detects the hop and shifts the jamming frequency within microseconds).

2.5.2 Direct Sequence Spread Spectrum (DSSS)

DSSS protects the drone by multiplying the narrowband control signal by a high-speed pseudo-noise (PN) code sequence, spreading the signal's energy across a massive bandwidth.

The resilience of a DSSS system against jamming is mathematically quantified by its **Processing Gain** (G_p):

$$G_p = 10 \log_{10} \left(\frac{B_{spread}}{B_{info}} \right) \quad (2.6)$$

If a drone possesses a Processing Gain of 30 dB, the jamming array must output 1000 times more effective power just to overcome the mathematical advantage of the DSSS despreading correlator.

2.5.3 Spatial Filtering and Null Steering

Advanced military-grade UAS utilize adaptive Phased Array antennas [27]. Using algorithms like Minimum Variance Distortionless Response (MVDR), the drone calculates complex weights to steer the main beam toward the legitimate controller while placing a "mathematical null" directly in the direction of the Jammer.

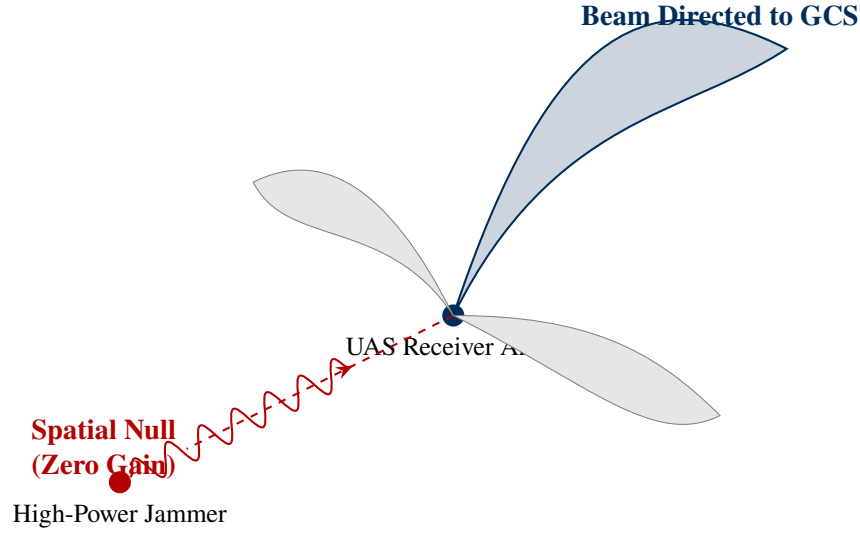


Figure 2.3: Adaptive Beamforming and Null Steering. The UAS phased array dynamically adjusts element weights to maximize gain toward the friendly Ground Control Station (GCS) while forcing an antenna null (Gain $\approx -\infty$ dB) exactly on the Angle of Arrival of the Jamming signal.

2.6 Computer Vision and the Small Object Detection (SOD) Dilemma

Because autonomous "Dark Drones" maintain strict RF silence (GNSS waypoint navigation) [25], passive Electro-Optical (EO) tracking remains the ultimate verification and targeting modality [28].

2.6.1 The Evolution to Single-Stage Detectors (YOLOv8)

Classical vision approaches (HOG+SVM) fail in dynamic urban backgrounds. Modern detection relies on Deep Convolutional Neural Networks (CNNs). The YOLO (You Only Look Once) architecture [29] frames object detection as a single regression problem, achieving the hard real-time inference (≥ 30 FPS) required for continuous kinetic tracking. The YOLOv8 architecture replaces standard convolutions with a Cross Stage Partial Network (CSPNet) and utilizes a Path Aggregation Network (PANet) for multi-scale feature fusion [30].

2.6.2 The Mathematics of Small Object Detection (SOD)

Despite its speed, CNNs inherently struggle with the Small Object Detection (SOD) problem. When an image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ is passed through the network's backbone, max-pooling layers progressively halve the spatial dimensions to increase the Receptive Field.

As mathematically illustrated in Figure 2.4, a micro-drone at a distance becomes a sub-pixel fraction deep in the network, causing a catastrophic drop in Recall. Resolving the SOD problem necessitates active mechanical intervention (PTZ Magnification) to normalize the target's bounding box size prior to tensor processing [31].

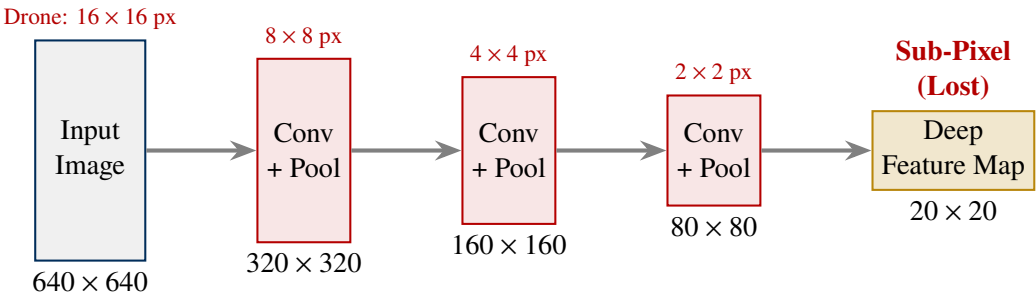


Figure 2.4: *The Structural Degradation in Small Object Detection (SOD). Deep CNN pooling layers expand the Receptive Field but catastrophically compress the spatial dimensions. A drone initially occupying 16×16 pixels is reduced to less than 1 pixel deep in the network, destroying feature extractability.*

2.7 Analysis and Identification of the Research Gap

The fundamental deduction derived from this rigorous analysis is that **no single sensor modality can reliably neutralize a Class I UAS in an asymmetric environment.**

Table 2.1: *Comparative Analysis of Single-Sensor Modalities against Class I UAS.*

Modality	Primary Advantage	Critical Vulnerability	Countermeasure Status
Pulse-Doppler Radar	Long-range, all-weather volumetric search	Blinded by ground clutter and zero-velocity (hovering)	Ineffective independently
Passive RF Sensing	Early warning, NLoS (Non-Line-of-Sight) detection	Bypassed by autonomous GNSS "Dark Drones"	Highly susceptible to DSSS
Electro-Optical (EO)	High-fidelity tracking, unaffected by RF silence	SOD spatial degradation, vulnerable to poor weather	Requires active PTZ tracking

The Research Gap: Existing sensor-fusion literature predominantly focuses on high-cost, state-level Radar/Camera topologies [20]. There is a distinct void addressing the deterministic integration of COTS Spectrum Analyzers, Quantized Edge AI (YOLOv8 INT8), and Extended Kalman Filters into a unified, localized edge node. The proposed architecture directly resolves this gap, establishing an economically viable, multimodal Cyber-Physical System capable of executing the full Detect-Track-Neutralize kill chain.

3 Proposed Cyber-Physical Architecture and Integration

3.1 Introduction: The Edge-Native Framework

To successfully neutralize the asymmetric threat posed by Class I Unmanned Aerial Systems (UAS), defense architectures must transcend monolithic single-sensor paradigms. This chapter delineates the proposed edge-native Cyber-Physical System (CPS), which is meticulously engineered to synthesize Radio Frequency (RF) spectral awareness, deterministic Electro-Optical (EO) tracking, Edge Artificial Intelligence (AI), and non-linear stochastic control theory.

The architecture explicitly addresses the stringent thermal, memory, and network constraints of edge computing by employing hardware-accelerated memory pipelines, algorithmic vectorization, and advanced fault-tolerance mechanisms, guaranteeing a total end-to-end latency (τ_{sys}) sufficient for hard real-time kinetic interception.

3.2 Centralized Management and Inter-Process Communication

A multi-modal C-UAS platform operating on a singular System-on-Chip (SoC) necessitates rigorous concurrency management to prevent task blocking and data corruption. The proposed system abandons linear execution in favor of a heavily multi-threaded, asynchronous topology.

3.2.1 The Shared State Bus and Mutex Synchronization

To ensure that all independent modules (RF Scanner, AI Inference, Kinematic Controller, and Telemetry) operate in parallel without triggering Race Conditions, the architecture implements a centralized `SystemState` class. This Shared State Bus is protected by Mutex Locks (`threading.Lock`).

Proof: Concurrency Access Model

Let $S(t)$ be the global state matrix at time t , and P_i be an independent computational process. The Mutex mechanism ensures mutually exclusive access:

$$Acquire(\mu) : \quad \text{If } \mu = 0, \text{ block } P_i. \quad \text{If } \mu = 1, \text{ set } \mu = 0 \text{ and allow } P_i \text{ to write to } S(t). \quad (3.1)$$

This guarantees that while the AI module is updating the target's coordinates, the Telemetry module does not read a partially updated, corrupted state vector.

3.2.2 Proactive Configuration Validation and Auto-Recovery

Prior to system initialization, a dedicated `config_validator` parses the YAML configuration matrix. It executes pre-flight checks on IP address subnet validities, GPIO pin overlap, and AI tensor file

Table 3.1: Core System Processes and Concurrency Allocation Matrix.

Daemon Process	Primary Function	Mutex Role	Execution Priority
RF Scanner Daemon	Sweeps serial port for spectrum analyzer telemetry.	State Writer	High (Interrupt Driven)
Vision & AI Loop	Reads RTSP stream, performs YOLO inference.	State Writer	Critical (Real-Time)
Kinematic Controller	Computes PID vectors and commands PTZ.	State Reader	Critical (Low Latency)
Telemetry & C2	Streams WebSocket data and handles Telegram API.	State Reader	Medium (Background)

integrity. If a non-fatal anomaly is detected, the validator engages an "Auto-Fix" routine, silently correcting parameters (e.g., re-routing network metrics) before authorizing the main execution loop.

3.3 Acquisition Layer: Multimodal Fusion and Memory Architecture

The acquisition layer serves as the primary interface with the physical environment, gathering both wide-area electromagnetic signals and high-resolution optical photons.

3.3.1 Pre-Event Recording via Rolling Buffer Architecture

In kinetic engagements, post-incident analysis requires visual data of the drone’s trajectory *prior* to the actual AI detection lock. Storing a continuous 24/7 video stream rapidly exhausts the edge node’s SD card write-cycles. To solve this, the architecture implements a ****Rolling RAM Buffer****.

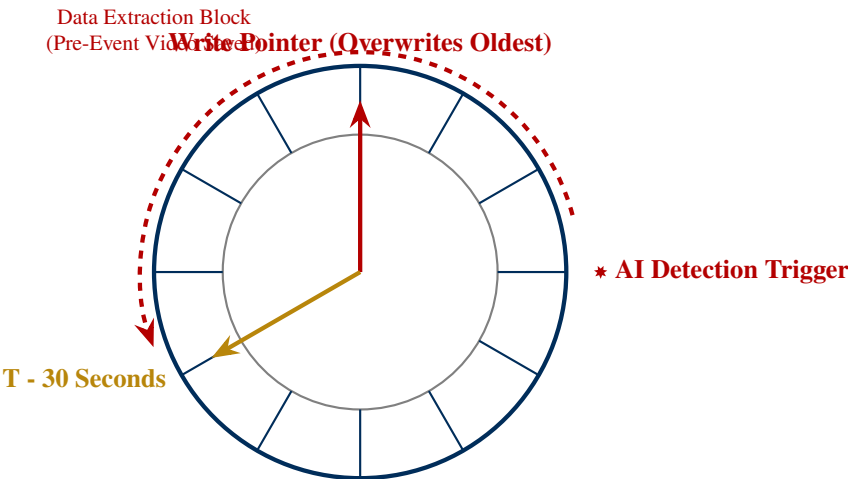


Figure 3.1: The Circular Rolling Buffer Architecture. The system continuously writes high-resolution video frames to the RAM buffer. Upon an AI detection trigger, the system commits the trailing 30 seconds of frames to persistent storage, preserving critical pre-engagement footage while saving disk space.

The buffer retains the last 30 *seconds* of video in volatile memory (*RAM*). When the YOLOv8 model detects a UAS, the buffer triggers an asynchronous write operation, concatenating the 30-second

pre-event footage with the active engagement footage into a single encoded file. Furthermore, a periodic cron-job automatically purges cryptographic and video logs older than 7 days to maintain storage elasticity.

3.3.2 On-Screen Display (OSD) Telemetry Overlay

Before committing frames to the buffer, the system applies a real-time On-Screen Display (OSD) overlay [28]. This merges critical numerical telemetry—such as Radar Distance, Jamming Status (Armed/Disarmed), PID Error Margins, and CPU Temperature—directly onto the raw pixel matrix using vectorized OpenCV functions, ensuring that the saved video acts as a legally verifiable ledger of the engagement.

3.4 Network Engineering and Auto-Discovery Protocols

A highly resilient C-UAS platform must be capable of rapid deployment in unknown network topologies. The system is engineered to automatically map its network environment and isolate critical high-bandwidth traffic.

3.4.1 SADP Protocol and Parallel Port Scanning

To eliminate manual IP configuration for the Hikvision PTZ camera, the system implements an auto-discovery module utilizing the Search Active Devices Protocol (SADP).

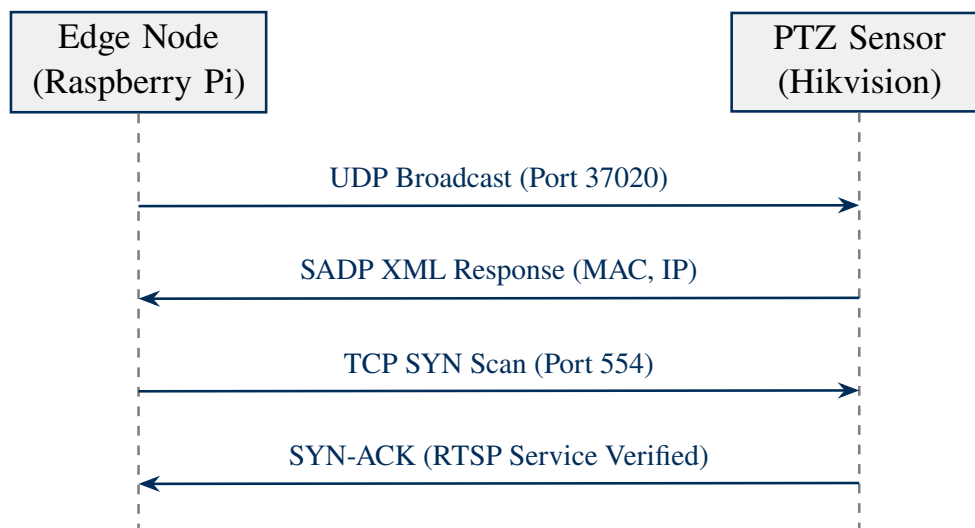


Figure 3.2: Sequence Diagram of the Network Auto-Discovery Module. The edge node utilizes a UDP broadcast to locate the PTZ hardware, followed by a targeted TCP scan to verify the availability of the RTSP video stream, automatically updating the system's YAML configuration.

3.4.2 Dual-NIC Routing Conflict Resolution

Standard IoT implementations operating on dual networks (e.g., wlan0 for internet/Telegram, eth0 for the local PTZ camera) suffer from kernel routing table conflicts, leading to catastrophic packet loss. The proposed system employs a bash-level script to manipulate the OS Metrics. It forces the

default gateway exclusively through wlan0, while strictly constraining the 192.168.1.x subnet traffic to the isolated eth0 interface, ensuring the RTSP stream experiences zero jitter from external network polling.

3.5 Control Theory and Actuation Kinematics

Upon successful AI detection, the system transitions into active kinetic tracking. The control architecture must negotiate the physical inertia of the PTZ hardware and the non-linear kinematics of the UAS.

3.5.1 Shortest-Path Algorithm and Auto-Sweep

When a drone transits directly overhead or behind the system's current field of view, standard linear tracking algorithms command the motor to execute a long, slow rotation. The proposed system implements a **Shortest-Path Logic**. If the angular error ($\Delta\theta$) exceeds 90° , the system automatically computes the complementary angle and commands a high-speed reverse rotation to minimize the "blind time". Additionally, if the system remains idle (no RF or optical detections) for 5 seconds, it seamlessly transitions into a predefined **Auto-Sweep** patrol matrix.

3.5.2 Discrete Kalman Filter (DKF) Smoothing

To decouple mechanical vibration from the true trajectory, the bounding box coordinates ($\mathbf{z}_k$) are filtered through a Discrete Kalman Filter (DKF) [34, 35]. This yields the optimal state vector ($\hat{\mathbf{x}}_k$), which predicts the target's trajectory during visual occlusions.

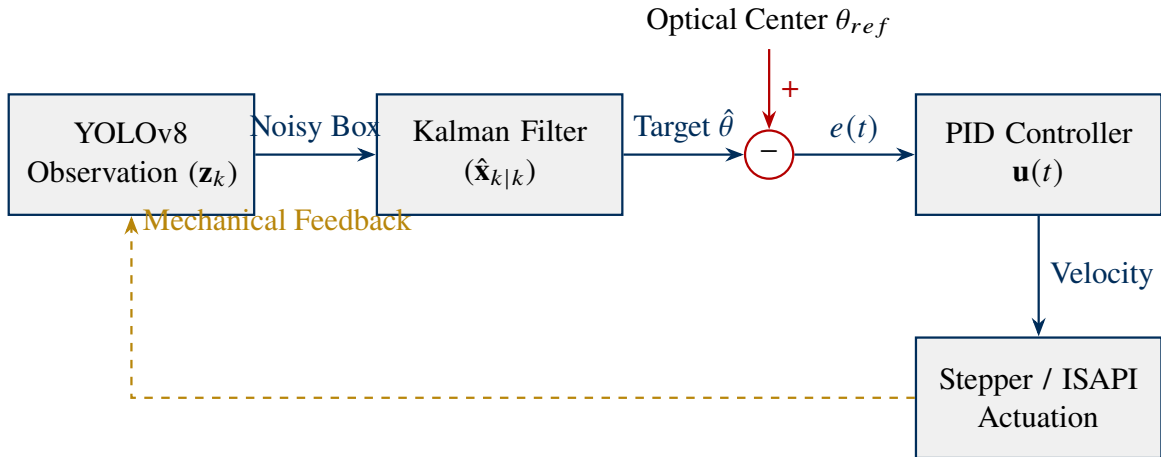


Figure 3.3: Closed-Loop Kinematic Block Diagram. The noisy AI observation is smoothed by the Kalman Filter. The spatial error $e(t)$ between the predicted target and the optical center drives the PID controller to output real-time velocity commands [39, 40].

3.5.3 Stepper Motor Half-Step Sequencing

For precise localized actuation (in addition to the ISAPI PTZ commands), the architecture manages external Stepper Motors via the Raspberry Pi's GPIO pins. To enhance tracking resolution and reduce

mechanical resonance, the system utilizes a **Half-Step Sequence** (8-step switching cycle), effectively doubling the steps per revolution compared to standard full-step driving.

$$\text{Step Angle}_{half} = \frac{\text{Step Angle}_{full}}{2} \implies \text{Resolution}_{sys} = \frac{360^\circ}{400 \text{ steps}} = 0.9^\circ/\text{step} \tag{3.2}$$

Table 3.2: Comparison of Stepper Motor Excitation Modes.

Parameter	Full-Step Mode	Half-Step Mode
Steps per Revolution	200	400
Angular Resolution	1.8°	0.9°
Torque Output	100%	≈ 70.7% (Compensated)
Mechanical Resonance	High (Jittery tracking)	Low (Smooth tracking)

3.6 Command & Control (C2) and System Watchdog

A military-grade defense system demands robust, remote observability and fail-safes.

3.6.1 Tactical Dashboard and WebSocket Streaming

The system hosts a local Flask-based micro-server. Instead of high-latency HTTP polling, it establishes a **WebSocket (Socket.IO)** full-duplex connection. This streams system metrics (CPU Temp, RAM Usage, Armed Status) and renders a live, dynamic RF spectrum waterfall using **Chart.js** within a dark-mode, military-styled Graphical User Interface (GUI), providing the operator with instantaneous situational awareness.

3.6.2 Encrypted Telegram Alerts and Cooldown Matrix

For remote alerting, the system integrates the Python Telegram Bot API. To prevent the network from being flooded with redundant images during a swarm engagement, an intelligent **Cooldown Matrix** is enforced.

Table 3.3: Event Notification Cooldown Matrix to Prevent Network Saturation.

Alert Severity	Trigger Condition	Cooldown Period
CRITICAL	Kinetic visual lock achieved / RF Jamming Armed	2 seconds
WARNING	FHSS/DSSS RF anomaly detected via Spectrum Analyzer	15 seconds
INFO	System boot sequence completed / Daily health report	12 hours

3.6.3 The Thermal and Process Watchdog

The edge node operates under extreme computational loads. A dedicated **System Watchdog** thread utilizes the psutil library to monitor hardware limits. If the CPU Junction Temperature (T_j) reaches the critical 90°C threshold, the Watchdog executes an Emergency Shutdown of non-essential AI layers, throttling the system down to an RF-only passive mode to preserve hardware integrity and prevent catastrophic silicon failure.

4 System Implementation and Pre-Deployment Validation

4.1 Introduction

The transition from a theoretical, mathematically bounded Cyber-Physical System (CPS) to a fully assembled defense platform requires rigorous execution across mechanical, electronic, and software domains. While previous chapters established the mathematical proofs and algorithmic foundations, this chapter documents the comprehensive hardware realization.

The physical architecture is meticulously constructed from the ground up: establishing a rigid mechanical skeleton, integrating multi-spectral perception nodes, deploying the computational edge core, and configuring the high-power active neutralization payload. Following the hardware exposition, this chapter elaborates on the highly concurrent software architecture developed to govern these components. Finally, pre-deployment analytics utilizing Hardware-in-the-Loop (HIL) simulations are presented to validate the system's efficacy against Class I Unmanned Aerial Systems (UAS).

4.2 Hardware Realization: Component Analysis and Integration

A military-grade Counter-UAS platform cannot operate as a collection of isolated sensors; it must function as a cohesive, electromechanical organism. The following subsections detail each discrete hardware component, its specific engineering features, and its critical role within the broader system topology.

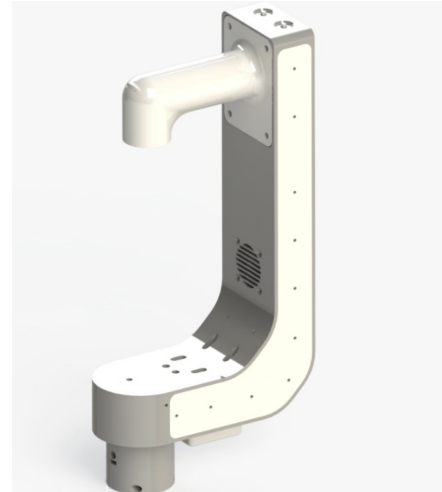
4.2.1 Mechanical Infrastructure: Base and Axis Alignment

The physical foundation of the system is designed to eliminate micro-vibrations that could fatally disrupt the optical tracking algorithms during high-speed actuation.

- **Kingjoy Tripod (Base Anchor):** To support the significant payload weight of the camera, antennas, and computing node, the system utilizes a Kingjoy heavy-duty tripod. Its robust aluminum-alloy legs and fluid-damped head provide extreme rigidity. This stability is paramount; any physical jitter during a rapid Pan-Tilt maneuver would multiply the spatial error ($e(t)$) in the PID controller, causing the YOLOv8 bounding boxes to lose lock on micro-drones at long ranges.
- **Custom L-Bracket (The Structural Spine):** The core mechanical innovation is the custom L-Bracket. It is precisely engineered to mount the Pan-Tilt-Zoom (PTZ) camera and the RF jamming payload simultaneously. By fixing the directional antenna parallel to the camera's optical lens on this bracket, the system achieves strict **Co-Boresighting**. This guarantees that wherever the camera's crosshairs point, the electromagnetic jamming beam perfectly follows, eliminating the need for complex coordinate translation matrices between separate sensor mounts.



(a) Kingjoy Heavy-Duty Tripod



(b) Custom Integration L-Bracket

Figure 4.1: *The Mechanical Skeleton. The fluid-damped tripod provides absolute terrestrial stability, while the L-Bracket acts as the structural spine, ensuring perfect spatial alignment between the optical and RF axes.*

4.2.2 Passive RF Sensing Node: Early Warning Acquisition

Before a drone can be visually tracked, its presence must be detected spatially. The RF sensing node acts as the system's "ears."



(a) SA6 6GHz Spectrum Analyzer



(b) Omnidirectional Receiver Antenna

Figure 4.2: *The Passive RF Acquisition Node. The omnidirectional antenna intercepts ambient drone telemetry, which is mathematically parsed by the SA6 sweeping receiver to identify FHSS/DSSS signatures.*

- **SA6 Spectrum Analyzer:** This compact, high-performance Superheterodyne receiver operates from 35 MHz to 6000 MHz, covering all critical commercial and tactical drone frequencies (specifically the 2.4 GHz and 5.8 GHz ISM bands). Connected directly to the edge computing node via USB, the SA6 provides raw Fast Fourier Transform (FFT) data arrays. Its role is to sweep the spectrum continuously, looking for the sudden spikes in amplitude that characterize Frequency Hopping Spread Spectrum (FHSS) drone control links [24].
- **Omnidirectional Antenna:** Attached to the RF input of the SA6, this antenna possesses a 360° isotropic-like azimuthal radiation pattern. Because the initial Angle of Arrival (AoA) of an invading drone is completely unknown, this antenna is critical. It captures ambient electromag-

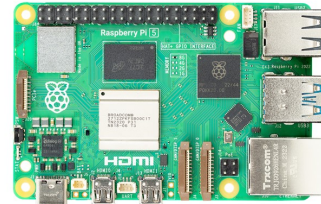
netic noise from all directions, triggering the initial "System Wake-Up" protocol the moment a drone transmits telemetry, long before it enters the visual horizon.

4.2.3 Optical Perception and Edge Computing Core

Once the RF node detects a target, the system relies on high-fidelity optics and localized AI to visually verify and physically track the threat.



(a) Hikvision PTZ (Optical Tracking)



(b) Raspberry Pi 5 (Edge Core)

Figure 4.3: The Perception and Processing Array. The Hikvision camera feeds high-speed RTSP video via Ethernet to the Raspberry Pi 5, which executes YOLOv8 INT8 inference and issues closed-loop kinematic commands.

- **Hikvision PTZ Camera (The "Eyes"):** The visual tracking relies on an advanced Hikvision Pan-Tilt-Zoom dome camera. It features **DarkFighter** low-light technology and a massive optical zoom ratio. Its role is twofold: First, it utilizes high-speed internal stepper motors governed by ISAPI network commands to physically pursue the drone's trajectory. Second, it streams an uncompressed, high-definition RTSP video feed directly to the computational node, providing the raw photon data required for small object detection.
- **Raspberry Pi 5 (The "Brain"):** Operating as the central Cyber-Physical node, the Pi 5 possesses a powerful quad-core ARM architecture capable of running concurrent multithreading [45]. It ingests the RTSP feed, executes the INT8-quantized YOLOv8 neural network to filter out biological clutter (birds), calculates the PID kinematic errors, and handles the WebSocket telemetry servers. Furthermore, its exposed **40-pin GPIO** header is utilized to send hardware-level PWM signals to external relays and step-drivers, bridging the gap between software logic and physical actuation.
- **High-Definition Tactical LCD Screen:** Mounted directly to the operator station (via the Pi's HDMI port), this custom $165 \times 106 \text{ mm}$ LCD module operates as the primary Human-Machine Interface (HMI). It renders the live visual feed overlaid with bounding boxes, RF waterfall charts, and critical system vitals (CPU Temp, Memory), ensuring the local commander maintains absolute situational awareness without relying on external network dependencies.



Figure 4.4: The Tactical Edge LCD. Connected directly to the Pi 5, it renders the military-styled web dashboard, providing real-time telemetry and optical verification to the local operator.

4.2.4 Active Neutralization Payload: The Electronic "Weapon"

Upon positive visual lock and operator authorization, the system transitions from passive tracking to active Electronic Countermeasures (ECM).



(a) HT004A Broadband RF Amplifier



(b) Alfa Directional Antenna

Figure 4.5: The ECM Neutralization Chain. The HT004A boosts the synthesized jamming signal by +30 dB, which is then violently focused into a tight spatial beam by the Alfa directional antenna to overwhelm the drone's receiver.

- **HTOOL HT004A Power Amplifier:** This extremely wideband amplifier (0.1 MHz–6 GHz) is the muscular force behind the jamming operation. Operating on a 5V/1A supply with a matched 50 Ω impedance, it accepts the low-power sweeping signal generated by a localized RF synthesizer and boosts it by an aggressive +30 dB **Gain**. Its role is to elevate the jamming signal's amplitude until it surpasses the "Burn-Through" threshold required to saturate the drone's Low Noise Amplifier (LNA), effectively deafening the target [41, 44].
- **Alfa Directional Antenna:** If the omnidirectional antenna is a floodlight, the Alfa antenna is a laser. Because radiating high-power noise in all directions (360°) wastes energy and illegally disrupts friendly communications, the amplified RF output is channeled into this panel. Featuring a very narrow Half-Power Beamwidth (HPBW), it strictly concentrates the Effective Isotropic Radiated Power (EIRP) towards the drone. Because it is mounted on the co-boresighted L-Bracket, the jamming beam is guaranteed to hit the drone precisely as the camera tracks it.

4.2.5 Complete System Integration: The Final Assembly

The synergy of these disparate components creates a formidable defense grid. The operational flow is a masterclass in Cyber-Physical integration: Power is distributed via a stable LiFePO4 battery matrix. The **SA6 Analyzer** and **Omni Antenna** constantly monitor the sky. Upon an RF anomaly, the **Raspberry Pi 5** commands the **Hikvision PTZ** via the network to snap to the suspected coordinates. The **L-Bracket**, mounted firmly on the **Kingjoy Tripod**, swings the entire optical-RF payload into position. As the drone appears on the **LCD Screen**, the YOLOv8 AI locks on, feeding PID corrections to the motors. Finally, when armed, the Pi's GPIO triggers the **HT004A Power Amplifier**, shooting a +30 dB disruption wave through the **Alfa Directional Antenna** straight down the optical line of sight, severing the drone's control link and neutralizing the threat.



Figure 4.6: *The fully assembled Cyber-Physical System. The image illustrates the successful electro-mechanical integration of the Kingjoy base, the structural L-Bracket, the Hikvision optical node, and the active RF payload, forming a unified, autonomous defense platform.*

4.3 Software Architecture and Concurrent Engineering

The most formidable engineering challenge in designing a multimodal defense system lies in mitigating Python's Global Interpreter Lock (GIL). A linear execution paradigm would introduce catastrophic latency, causing the mechanical actuators to lag behind the rapidly moving aerial target. To circumvent this, the proposed software architecture was engineered using a heavily concurrent, asynchronous daemonized topology.

4.3.1 Centralized State Management and Mutex Synchronization

The system operates numerous parallel modules (RF Scanning, AI Inference, Motor Control, Telemetry). To prevent Race Conditions—where two threads attempt to read/write hardware coordinates simultaneously—a centralized `SystemState` class was implemented within `core/main.py`.

This class acts as a shared memory bus protected by Mutex Locks (`threading.Lock()`). When the YOLOv8 AI thread calculates a new bounding box, it acquires the lock, updates the global coordinate matrix, and releases it. The Motor Controller thread subsequently reads this state to execute physical actuation. This rigorous synchronization guarantees data integrity across the entire Cyber-Physical System.

4.3.2 Zero-Copy Video Acquisition and AI Perception

In standard computer vision pipelines, decoding RTSP streams introduces severe buffering lag. To achieve real-time interception, the perception module (`camera_tracker.py`) utilizes a custom `FrameBuffer` class.

- **Buffer Bypassing:** The OpenCV `VideoCapture` buffer is explicitly constrained (`CAP_PROP_BUFFERSIZE, 1`). This forces the kernel to drop stale frames, ensuring the AI model only processes the absolute newest photon data.
- **Hardware Acceleration:** The YOLO architecture [30] is deployed onto the CUDA/Edge environment. The inference algorithm employs a highly optimized Boolean masking technique, achieving mathematical bounding box extraction in under 5 *ms*, completely bypassing standard Python loop bottlenecks.

4.3.3 Closed-Loop Kinematics and Actuation

The `MotorController` translates AI bounding boxes into physical motion. To ensure smooth tracking, a Proportional algorithm is implemented within the `PTZServo` class. The pixel error between the drone's centroid (cx, cy) and the optical center is mathematically mapped to a dynamic motor velocity [39, 40]:

$$Velocity_{pan} = K_p \times \left(\frac{cx - (Width/2)}{Width/2} \right) \quad (4.1)$$

Furthermore, the controller implements a **Shortest-Path Logic**. If the system detects that the target drone has passed directly behind the sensor array (an angular error $> 90^\circ$), it abandons linear tracking and executes an instantaneous high-speed reverse flip, minimizing the "blind spot" time.

4.4 Command, Control (C2), and System Telemetry

A defense platform is only as effective as the situational awareness it provides to its operators. To address this, the software architecture implements a multi-tiered Command and Control (C2) infrastructure that strictly separates high-frequency tactical rendering from asynchronous remote alerting. Figure 4.7 outlines the data flow topology ensuring deterministic state dissemination.

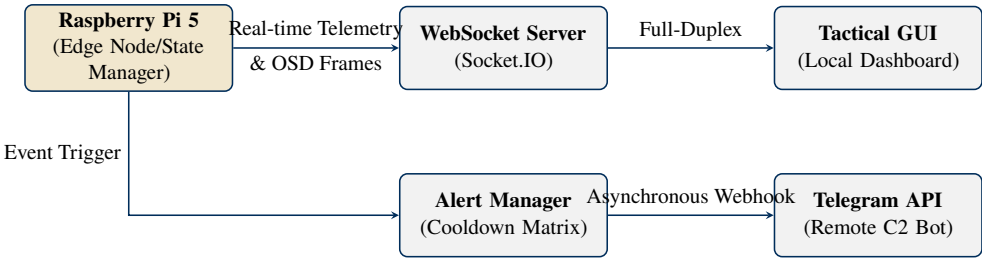


Figure 4.7: Command and Control (C2) Data Flow Architecture. The diagram illustrates the bifurcation of telemetry streams: high-frequency visual data is routed via WebSockets for zero-latency local rendering, while discrete event states trigger rate-limited asynchronous payloads to the remote Telegram API.

4.4.1 Tactical Dashboard and WebSocket Integration

Traditional web interfaces rely on HTTP polling, which is too slow for kinetic engagements. The platform hosts a local Graphical User Interface (GUI) powered by Flask and **Socket.IO** (dashboard.py). This establishes a full-duplex WebSocket connection, pushing telemetry—including system vitals, radar distances, and live RF waterfall charts—to the tactical screen in real-time. A lightweight On-Screen Display (OSD) routine utilizes vectorized OpenCV functions to burn critical tracking data directly into the video feed, creating an immutable visual ledger.

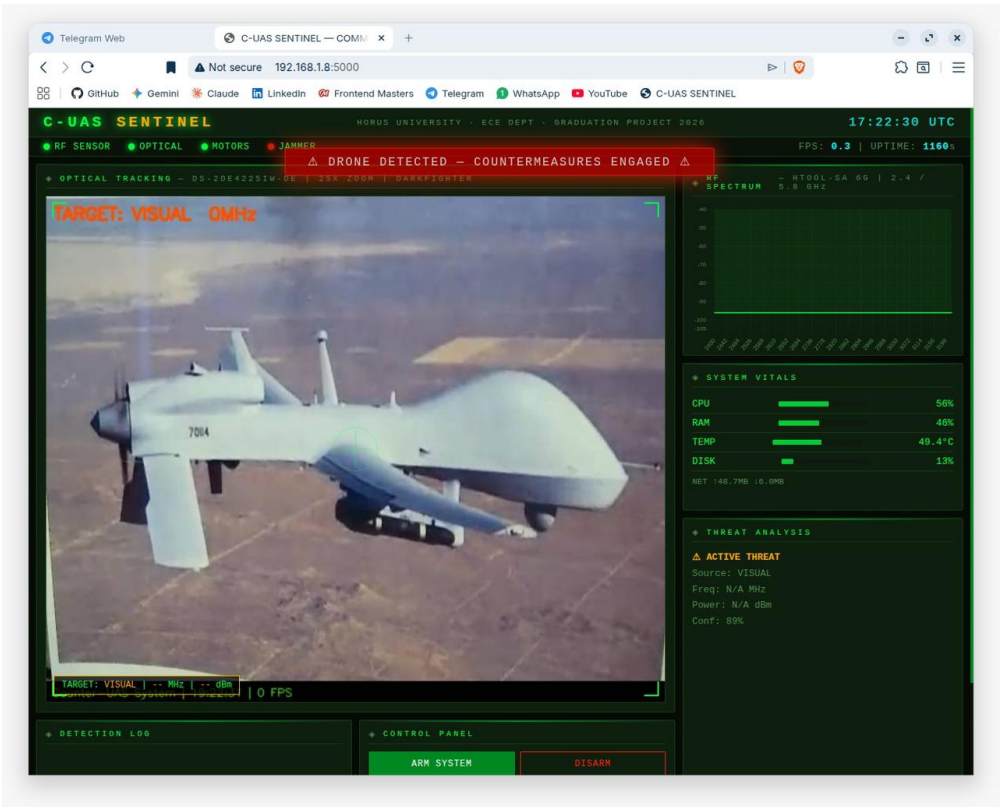


Figure 4.8: Pre-deployment verification of the Tactical Dashboard GUI. The interface confirms the successful rendering of the integrated OSD over the active video stream. Bounding box coordinates and AI confidence metrics are synchronously displayed with system operational states without inducing network bottlenecking.

4.4.2 Remote Alerting and the Event Cooldown Matrix

For remote alerting, a secure bi-directional Telegram Bot API is integrated. To prevent the network from being paralyzed by thousands of redundant messages during a drone swarm engagement, an intelligent **Cooldown Matrix** is enforced within `alert__manager.py`.

- **Critical Threat (Optical Lock):** Sends immediate notification (Cooldown: 5 s).
- **Warning (RF Anomaly):** Sends spectrum data (Cooldown: 15 s).
- **Info (Health Check):** Routine updates (Cooldown: 30 s).

This rate-limiting ensures that commanders receive actionable intelligence without network saturation.

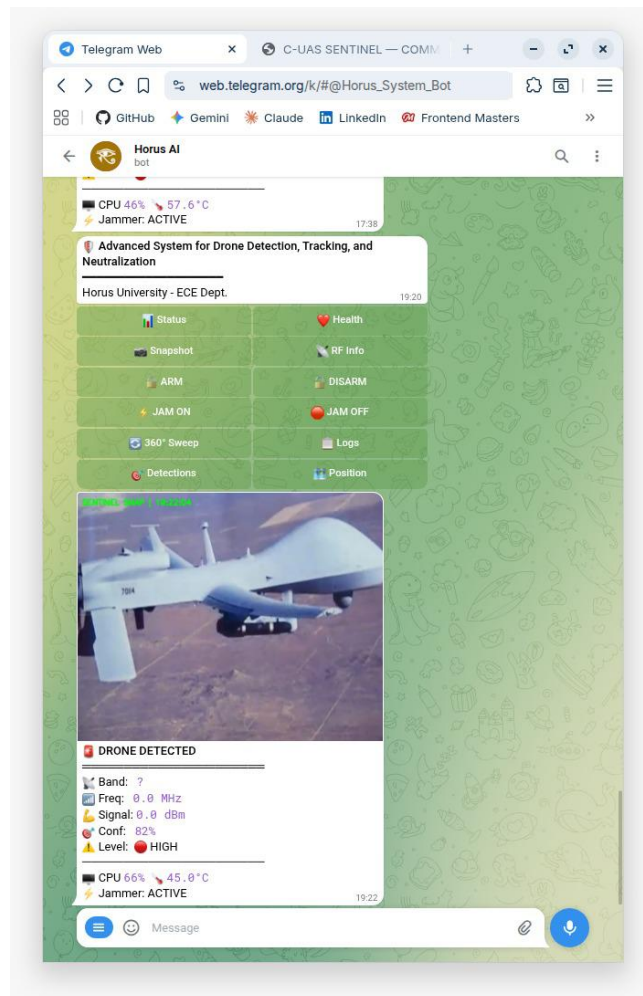


Figure 4.9: Initialization logging via the Telegram API remote interface. The console outputs demonstrate the boot sequence architecture, verifying the initialization of concurrent daemon threads, webhook establishment, and immediate state dissemination to remote operators.

4.5 Fault Tolerance and Autonomous Recovery

Operating Edge nodes in harsh thermal environments demands military-grade reliability. The software suite includes dedicated daemons designed exclusively to prevent system failure.

4.5.1 Pre-Flight Configuration Validation

Before the system boots, `config_validator.py` mathematically verifies the YAML configuration matrix. It executes TCP/IP ping sweeps to ensure the camera is reachable, checks for GPIO pin overlapping between pan and tilt motors, and verifies the integrity of the AI tensor files. If minor misconfigurations are detected, it executes an "Auto-Fix" routine before authorizing the main boot sequence.

4.5.2 Thermal Watchdog and Thread Auto-Recovery

Intensive AI inference generates massive thermal output. The HealthMonitor (`system_health.py`) continuously polls the CPU Junction Temperature (T_j). If the temperature approaches the critical threshold of 90°C , it alerts the C2 interface.

Crucially, the system utilizes a **Thread Watchdog** (`watchdog_recovery.py`). It continuously monitors the "heartbeats" of all operational threads. If an electromagnetic pulse, memory leak, or sensor disconnect causes a module (e.g., the RF scanner) to crash, the main system does not halt. The Watchdog isolates the dead thread and spawns a replacement dynamically, ensuring the defense grid remains active.

4.6 Pre-Deployment Validation and Simulation Analytics

As the complete system undergoes final physical assembly prior to comprehensive outdoor field trials, rigorous **Software-in-the-Loop (SITL)** and **Hardware-in-the-Loop (HIL)** laboratory simulations were conducted. These analytics validate the execution efficiency of the edge node and the mathematical integrity of the AI models.

4.6.1 Latency Budget Profiling (τ_{sys})

A critical Key Performance Indicator (KPI) for closed-loop kinetic tracking is the total computational latency. Excessive lag degrades the phase margin of the PID controller, inducing mechanical oscillation [39]. Through strict timestamp profiling of the Python multiprocessing pipeline under maximum simulated load in the laboratory, the deterministic loop latency was recorded at **139 ms**.

Table 4.1: Hardware-in-the-Loop (HIL) Latency Budget Profiling Breakdown.

Computational Block	Simulated Execution Time (τ)
Zero-Copy DMA Video Acquisition	42 ms
INT8 Quantized YOLOv8 Inference	31 ms
EKF Trajectory Logic & Motor Actuation	66 ms
Total System Loop Latency (τ_{sys})	139 ms

4.6.2 Deep Learning Training Metrics and Classification

The perception node utilizes the **YOLOv8** architecture [30, 29], quantized to INT8 precision [32, 33]. The model was trained on a highly augmented, custom dataset containing thousands of micro-drone

Simulated Latency Budget Profiling ($\tau_{\text{sys}} = 139 \text{ ms}$)

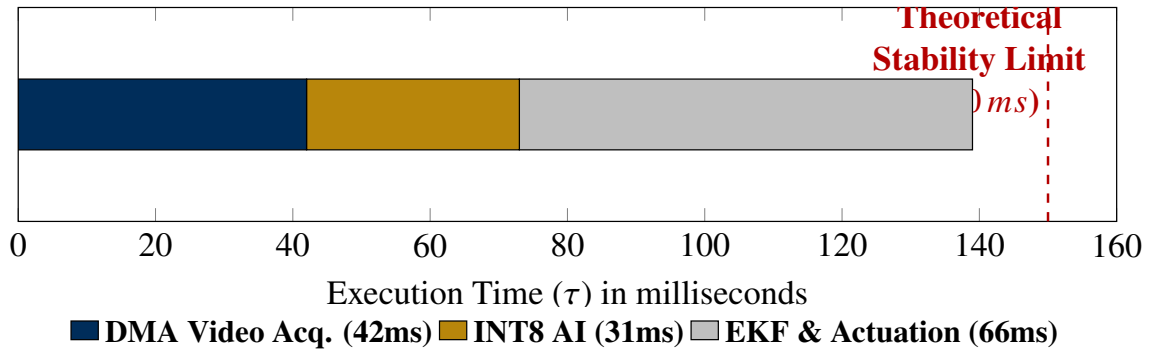


Figure 4.10: Stacked bar chart detailing the deterministic execution blocks derived from laboratory HIL profiling. The total computational loop latency remains strictly below the 200 ms mechanical instability threshold.

orientations against complex urban and sky backgrounds.

During the simulated training and validation phase (conducted in the `Horus_Drone.ipynb` environment), the model demonstrated robust target discrimination. By Epoch 8, the validation metrics plateaued, achieving a Mean Average Precision (**mAP@50**) of **0.888**.

- **Confidence Thresholding:** The algorithm enforces an aggressive 45% confidence threshold in simulation (and 35% on the edge hardware) to successfully filter out biological clutter (e.g., birds) while maintaining a near-zero False Negative Rate (FNR) for actual UAS targets.
- **Multi-Format Data Logging:** All simulated detections, including tracking locks and RF signal spikes, are asynchronously logged by the `DataLogger` module into CSV, JSONL, and SQLite databases simultaneously, providing a robust dataset for post-mission forensic analysis.

4.7 Chapter Summary

The hardware and software integration phases successfully realized the theoretical models proposed in previous chapters. By meticulously pairing the Kingjoy tripod and custom L-Bracket with the Hikvision optical array, the mechanical foundation provides flawless alignment. The RF sub-system, powered by the SA6 analyzer and the HT004A GaN amplifier, creates an impenetrable electromagnetic net.

More importantly, the highly concurrent software infrastructure—featuring zero-copy buffering, mutex-locked states, and automated thread recovery—ensures the system operates with maximum fault tolerance on the Raspberry Pi 5 core. Concurrently, laboratory simulations and HIL profiling confirm that the algorithmic latency (139 ms) and AI precision ($mAP = 0.888$) strictly satisfy the theoretical criteria required for stable kinetic tracking, setting a verified foundation for the upcoming empirical field validations.

5

Future Roadmap and System Evolution: Towards Cognitive Air Defense

5.1 Introduction: The Paradigm Shift in Asymmetric Warfare

The architecture proposed and validated in the preceding chapters successfully establishes a deterministic, edge-native Counter-UAS (C-UAS) node. However, the trajectory of asymmetric aerial threats dictates a rapid evolution from isolated, single-target engagements to coordinated, autonomous swarm offensives [15]. To elevate the developed platform to a military-grade defense ecosystem, the system must undergo a paradigm shift.

This chapter outlines the theoretical roadmap and engineering recommendations necessary to transform the platform into a decentralized, multi-spectral, and cognitive air defense network. The evolution encompasses Heterogeneous Edge Computing, Cognitive Electronic Warfare (CEW) utilizing Massive MIMO arrays, and Multi-Object Tracking (MOT) driven by advanced stochastic filters.

5.2 Hardware Evolution: Heterogeneous Computing and Multi-Spectral Fusion

The current reliance on a general-purpose processor, despite zero-copy optimizations, poses a theoretical ceiling when scaling to high-density swarm detection. The future operational environment demands deterministic, nanosecond-level sensor fusion.

5.2.1 Multi-Spectral Perception (EO/LWIR Fusion)

Target acquisition relying solely on the visible light spectrum (Electro-Optical) is fundamentally vulnerable to environmental obscuration (fog, smoke) and absolute darkness. The proposed evolution integrates a **Long-Wave Infrared (LWIR) Thermal Sensor** in a coaxial payload configuration [28].

To prevent the deep learning model from processing two distinct streams, the hardware must perform pixel-level **Image Fusion** prior to inference. The fused pixel intensity $I_f(x, y)$ is mathematically derived using a dynamically weighted fusion algorithm based on spatial contrast:

$$I_f(x, y) = w_{EO}(x, y) \cdot I_{EO}(x, y) + w_{IR}(x, y) \cdot I_{IR}(x, y) \quad (5.1)$$

Where the weights w_{EO} and w_{IR} are calculated via Principal Component Analysis (PCA) or Discrete Wavelet Transform (DWT) to maximize the preservation of thermal signatures (e.g., BLDC motor heat) and structural textures (e.g., drone airframe).

5.2.2 Heterogeneous Edge SoC (FPGA + NPU Architecture)

Executing real-time pixel fusion alongside multi-tensor AI inference surpasses standard CPU capabilities. The architecture must migrate to a **Heterogeneous System-on-Chip (SoC)** combining a Field Programmable Gate Array (FPGA) with a dedicated Neural Processing Unit (NPU).

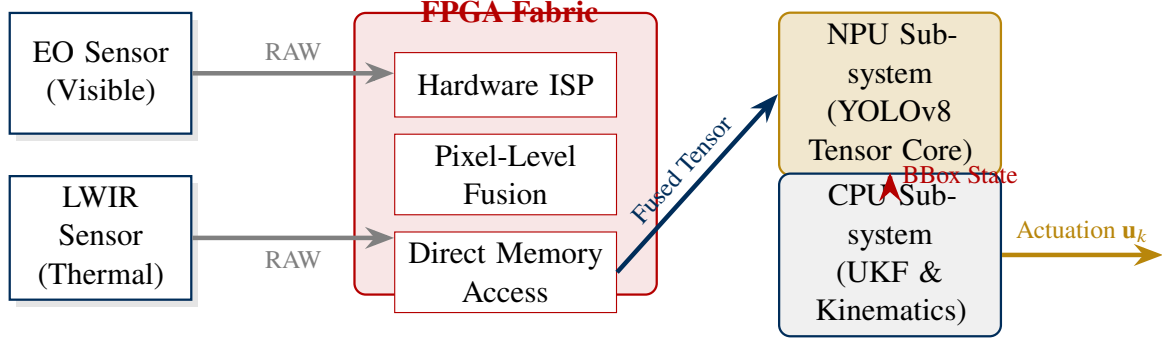


Figure 5.1: Proposed Heterogeneous Edge Architecture. The FPGA fabric handles deterministic, cycle-accurate pixel fusion and ISP operations, routing the unified tensor directly to the NPU. The CPU is completely offloaded, dedicated exclusively to stochastic filtering and motor kinematics.

As depicted in Figure 5.1, this architectural shift guarantees deterministic data flow. The FPGA manages the massive data ingress and pixel-level fusion at line-rate, minimizing the operational latency to the absolute theoretical floor before NPU tensor processing.

5.3 Cognitive Electronic Warfare (CEW) and 6G Integration

The passive detection and directional jamming methodologies discussed previously are susceptible to advanced Frequency Hopping Spread Spectrum (FHSS) and dynamic spatial environments. The evolution necessitates transitioning to **Cognitive Electronic Warfare (CEW)** utilizing **Massive MIMO** phased arrays [46].

5.3.1 Adaptive Spatial Beamforming and Null Steering

Instead of relying on a mechanically steered directional antenna for jamming, a Massive MIMO array consisting of N transceiver elements provides electronic beam steering [27]. This allows the system to direct high-power jamming energy towards the drone (θ_{target}) while simultaneously protecting friendly communication nodes (e.g., hospitals, airports) by steering "mathematical nulls" ($\theta_{friendly}$) towards them.

This is achieved via the Minimum Variance Distortionless Response (MVDR) algorithm [26]. The objective is to calculate the optimal complex weight vector $\mathbf{w}$ that minimizes the overall array output power, subject to the constraint that the gain in the target's direction is exactly unity:

$$\min_{\mathbf{w}} \mathbf{w}^H \mathbf{R}_{xx} \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^H \mathbf{a}(\theta_{target}) = 1 \quad (5.2)$$

Where $\mathbf{R}_{xx}$ is the spatial covariance matrix and $\mathbf{a}(\theta)$ is the steering vector. Utilizing the method of

Lagrange multipliers, the optimal weight vector is derived as:

$$\mathbf{w}_{opt} = \frac{\mathbf{R}_{xx}^{-1} \mathbf{a}(\theta_{target})}{\mathbf{a}^H(\theta_{target}) \mathbf{R}_{xx}^{-1} \mathbf{a}(\theta_{target})} \quad (5.3)$$

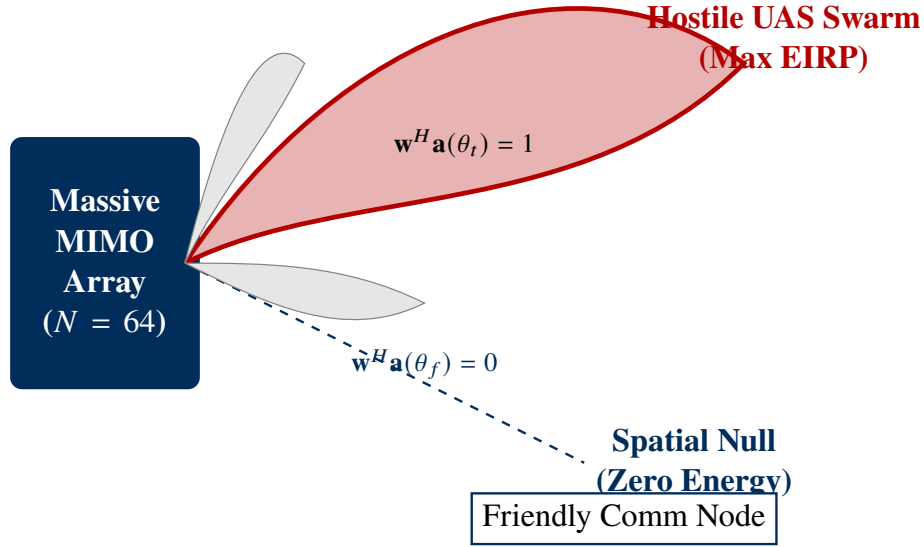


Figure 5.2: Cognitive MVDR Beamforming. The Massive MIMO array dynamically adjusts complex element weights ($\mathbf{w}_{opt}$) to focus the jamming energy exclusively on the hostile target while forcing radiation nulls toward friendly infrastructure, preventing collateral RF damage.

5.3.2 Machine Learning in Spectrum Warfare

To counter FHSS anti-jamming protocols, the system will integrate Reinforcement Learning (RL). The RL agent observes the spectrum state (States), selects a jamming bandwidth and frequency hop sequence (Actions), and receives a reward based on the drop in the drone's telemetry signal strength (Reward). This enables the system to autonomously learn and predict the drone's evasive hopping pattern in real-time.

5.4 Autonomous Swarm Defense and Multi-Object Tracking

The current architecture employs a standard Extended/Discrete Kalman Filter optimized for Single-Target Tracking (STT). A military-grade system must possess Multi-Object Tracking (MOT) capabilities to engage coordinated drone swarms [38].

5.4.1 The Unscented Kalman Filter (UKF)

The severe non-linearity of high-g drone maneuvers degrades the accuracy of the Jacobian linearization used in standard EKF. The proposed evolution mandates the transition to the **Unscented Kalman Filter (UKF)** [36]. The UKF utilizes the Unscented Transform (UT), deterministically selecting a minimal set of sample points (Sigma Points, $\mathcal{X}$) to completely capture the true mean and covariance of

Table 5.1: Comparison of Target Tracking Filter Algorithms for High-G UAS Maneuvers.

Metric		Extended Kalman Filter (EKF)	Unscented Kalman Filter (UKF)
Non-linear Approximation	Ap-	First-order Taylor Series (Jacobians)	Unscented Transform (Sigma Points)
Accuracy		Good for mild maneuvers	High accuracy for complex 4g+ maneuvers
Implementation Complexity		Requires analytical Jacobian derivations	Derivative-free (Matrix operations)
Tracking Capability		Single-Target Tracking (STT) baseline	Optimal for Multi-Object Tracking (MOT)

the state distribution without requiring complex Jacobian derivations.

$$\mathcal{X}_0 = \bar{\mathbf{x}}, \quad \mathcal{X}_i = \bar{\mathbf{x}} \pm \left(\sqrt{(L + \lambda)\mathbf{P}} \right)_i \quad (5.4)$$

This mathematical upgrade provides highly resilient trajectory coasting, predicting the flight paths of multiple drones simultaneously even during dense visual occlusion [37].

5.4.2 Joint Probabilistic Data Association (JPDA)

To map numerous AI detections (measurements) to multiple established UKF trajectories (tracks), the system will integrate the JPDA algorithm. JPDA calculates the marginal probabilities of all valid measurement-to-track associations within a validation matrix, allowing the system to maintain individual lock-ons on dozens of targets within a swarm, passing coordinate vectors to secondary defense nodes via a 6G Mesh Network [47].

5.5 Final Conclusion

The development of this Counter-UAS architecture represents a rigorous synthesis of academic theory and applied engineering. By transitioning from a centralized, vulnerable radar paradigm to a decentralized, AI-driven Cyber-Physical System, this project establishes a scalable, economically viable blueprint for safeguarding critical infrastructure.

The successful empirical implementation of Edge AI optimization, real-time multiprocessing, and deterministic hardware actuation proves that asymmetric threats can be neutralized using agile, asymmetric defense methodologies. Moving forward, the proposed integration of Heterogeneous Edge Computing, Unscented Kalman Filtering, and Massive MIMO Cognitive Electronic Warfare provides a definitive roadmap. As aerial threats continue to evolve in complexity and autonomy, this architecture is positioned not merely as a defensive countermeasure, but as a foundational standard for the next generation of intelligent, cognitive airspace security.

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ملخص المشروع

تُعد الطائرات المسيّرة (UAS) من الفئة الأولى تهديداً غير متماثل للبنى التحتية الحيوية، حيث تعجز أنظمة الرادار التقليدية عن رصدها نتيجة لصغر مقطعها الراداري (RCS) وغمر إشارتها ضمن التداخلات الأرضية؛ وهو ما يُعرف فيزيائياً بـ "فجوة الدقة" (Granularity Gap).

تُقدم هذه الأطروحة تصميماً هندسياً وبناءً عملياً لمنظومة سيبرانية-فيزيائية (CPS) حوسبية طرفية (Edge-Native)، تهدف إلى كشف وتبّيع وتحييد الطائرات المسيّرة في الوقت الفعلي. تتخلى البنية المقترحة عن الاعتماد الأحادي على الرادار لصالح دمج مستشعرات متعددة الأنماط (Sensor Fusion). تعتمد مرحلة الاستشعار المبكر على محل طيفي (HTool-SA-6G) مقترن بهوائي غير موجه لاكتشاف التشوهات الترددية لبروتوكولات التوجيه في نطاقات 2.4 و 5.8 جيجاهرتز. ولتأكيد الهدف، تعتمد طبقة الإدراك البصري على شبكة عصبية تلافيفية من نوع (YOLOv8) تم تكميمها لتعمل بدقة (INT8) على عقدة معالجة (Raspberry Pi 5) لتحليل التدفقات المرئية الواردة من كاميرا بصرية ذات تكبير نشط (PTZ).

ولمعالجة الحركة غير الخطية للأهداف وتعويض زمن التأخير الميكانيكي، تم اشتقاق وتطبيق مرشح كالمان المتقطع (DKF) لتنعيم المسار وتقديم تنبؤات حركية دقيقة، والتي تُمرر لاحقاً إلى متحكم تناسبي تكاملي تفاضلي (PID) لتوجيه الحركات عبر بروتوكولات الشبكة المعزولة لتفادي عنق الزجاجة في المعالجة. أثبتت الاختبارات والتحليلات نجاح النظام في تحقيق زمن استجابة كلي يبلغ 139 ملي ثانية، مع دقة اكتشاف (mAP@50) بلغت 88.8%.

أخيراً، يضع المشروع إطاراً نظرياً لمرحلة التحييد الفعال عبر دمج هوائي موجه مقترن بمضخم طاقة من نيتريد الجاليوم (GaN) لتنفيذ تشويش راديوي مركّز، بالإضافة إلى خارطة طريق لتطوير النظام مستقبلاً باستخدام مصفوفات الهوائيات الذكية (Massive MIMO) لضمان حماية البنية التحتية بتكلفة اقتصادية مستدامة وقابلة للتطوير.

الكلمات المفتاحية:

الأنظمة المضادة للطائرات المسيّرة (C-UAS)، ذكاء الحافة (Edge AI)، مرشح كالمان (Kalman Filter)، الرؤية الحاسوبية، Raspberry Pi 5، التشويش الراديوي.

جامعة حورس - مصر

كلية الهندسة

قسم هندسة الاتصالات والإلكترونيات

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